Bhaskar Dwivedi

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PROFESSIONAL SUMMARY

AI Product Engineer and Computer Science undergraduate who builds intelligent applications end to end, from data pipelines and models to the APIs and interfaces users rely on. Experienced across machine learning, deep learning, and NLP, from a hybrid deep learning model for clinical risk prediction to an NLP pipeline trained on 400,000+ text pairs, backed by full-stack engineering and AWS certifications.

TECHNICAL SKILLS

AI / ML: Machine Learning, Deep Learning, Natural Language Processing, Autoencoders, Deep Neural Networks, Feature Engineering, Hyperparameter Optimization (PSO), Model Evaluation
Generative AI: Foundation Model Concepts, Prompt Engineering, Responsible AI Practices
ML Libraries: Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib
Product Engineering: React.js, Node.js, Express.js, REST APIs, Socket.io, JWT Authentication, MVC Architecture
Data & Cloud: PostgreSQL, MySQL, MongoDB, Data Modeling, AWS, Linux
Languages & Tools: Python, C++, JavaScript, SQL, Git, GitHub, Postman, Figma

PROJECT-BASED LEARNING

Applied Machine Learning & NLP Program, Centre of Excellence, HP–Intel AI Skills Lab (KIET) June 2026
Machine Learning, NLP, Python

- Completed a six-week, project-based program in Machine Learning and NLP, guided by industry mentors from Intel.
- Delivered a capstone NLP system end to end across data preprocessing, text cleaning, feature engineering, model training, and evaluation, using structured experiments to select models on quantitative metrics.

PROJECTS

Hybrid Deep Learning Model for Cardiovascular Disease Prediction October 2026
Python, Deep Learning, Autoencoder, DNN, Particle Swarm Optimization, Scikit-learn

- Engineered a hybrid deep learning pipeline that pairs an **Autoencoder** for unsupervised feature compression with a **Deep Neural Network** classifier to predict cardiovascular disease risk from clinical patient data.
- Integrated **Particle Swarm Optimization (PSO)** as a metaheuristic optimizer to tune the network configuration, replacing manual trial-and-error hyperparameter search.
- Built a reproducible preprocessing workflow for clinical attributes: missing-value handling, encoding, and scaling.
- Evaluated on Accuracy, Precision, Recall, F1-Score, and ROC-AUC, prioritizing recall to reduce missed high-risk cases.

Duplicate Question Detection using NLP and Machine Learning June 2026
Python, NLP, TF-IDF, Scikit-learn, XGBoost, Pandas, NumPy

- Developed a semantic similarity classifier on the **Quora Question Pairs** dataset (**400,000+** pairs) to flag duplicate questions, a core capability for deduplicating Q&A and support content.
- Designed an NLP pipeline with text normalization, tokenization, and **TF-IDF** vectorization; trained and benchmarked **Logistic Regression**, **Random Forest**, and **XGBoost**, selecting the best model using Accuracy, Precision, Recall, F1-Score, and confusion matrices.
- Achieved **76% test accuracy** by iterating on text preprocessing and feature engineering.

Smart Task Management System July 2026
React.js, Node.js, Express.js, MongoDB, Socket.io, JWT

- Built a full-stack task management platform with JWT authentication, role-based access control, and RESTful APIs on an MVC architecture backed by an optimized MongoDB schema.
- Implemented real-time updates and notifications with Socket.io, reducing status-update latency for end users.

EDUCATION

B.Tech in Information Technology, KIET Group of Institutions, Ghaziabad 2023 – 2027
Senior Secondary (Class XII), Tathagat Gyansthli Senior Secondary School (CBSE) 2021 – 2022
89%, Mathematics & Computer Science
Secondary (Class X), Tathagat Gyansthli Senior Secondary School (CBSE) 2019 – 2020
94%

CERTIFICATIONS

- **AWS Certified AI Practitioner** — Amazon Web Services
- **AWS Certified Cloud Practitioner** — Amazon Web Services

ACHIEVEMENTS

- **Runner-Up**, IEEE SSH24 (Phase 1) — National-Level Hackathon
- **2nd Place**, HacktheChain 2.0 — National-Level Hackathon, IIIT Kota
- Solved **300+** problems on LeetCode across data structures and algorithms